

The following questions are intended to guide you through the most important concepts that you must understand from the lectures and the material provided.

A good exercise would be to study the lectures/course material until you are able to answer them without having to read or search for it in the slides.

Lecture 4: Biometric modalities

Questions: fingerprints

- Ways of acquiring fingerprints (slides 7-13)
- Anatomy of fingerprints: macro-points (loop, deltas), micro-points (minutiae), and type of fingerprints based on the arrangement of macro-points (slides 14-15)
- Typical steps in fingerprint recognition: orientation extraction (slide 18), frequency (19), enhancement (20) and comparison (22)
- Main challenges (slides 23-24), using slides 25-29 as support to see the effects of different perturbations on the images and the scores

Questions: face

- Advantages over another modalities (slide 37-39) but why face recognition (FR) is more difficult (slide 40)
- Example of face recognition using local landmarks (slide 44) and global (holistic) recognition (slide 45)

Questions: iris

- Understand the impact of light on iris recognition and why near-infrared (NIR) acquisition is preferred (slides 55-57)
- Typical steps in iris recognition: segmentation (slide 59), normalization to a rectangle (61) and filtering (58)
- Factors that can negatively affect iris recognition (slide 63)

Questions: other biometrics

- Which other biometric modalities are in use: palmprint and DNA, mainly for police investigation (slide 72) and other commercial/laboratory modalities (slides 73-76)
- What is soft-biometrics and which is its utility in biometric recognition (slide 77)